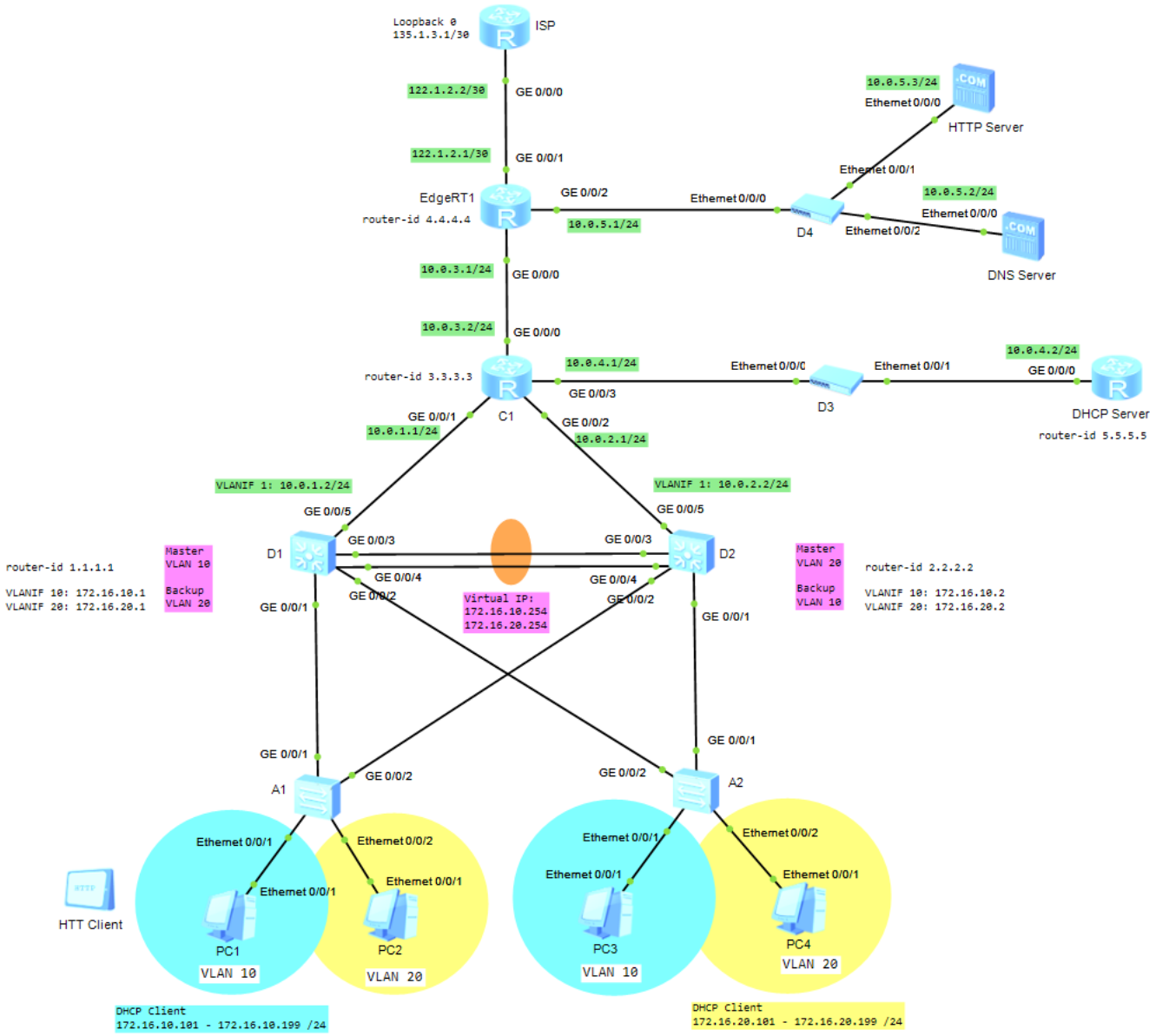
ЗЕРТХАНАЛЫҚ ЖҰМЫСТЫ орындауға арналған ӘДІСТЕМЕЛІК НҰСҚАУЛЫҚ

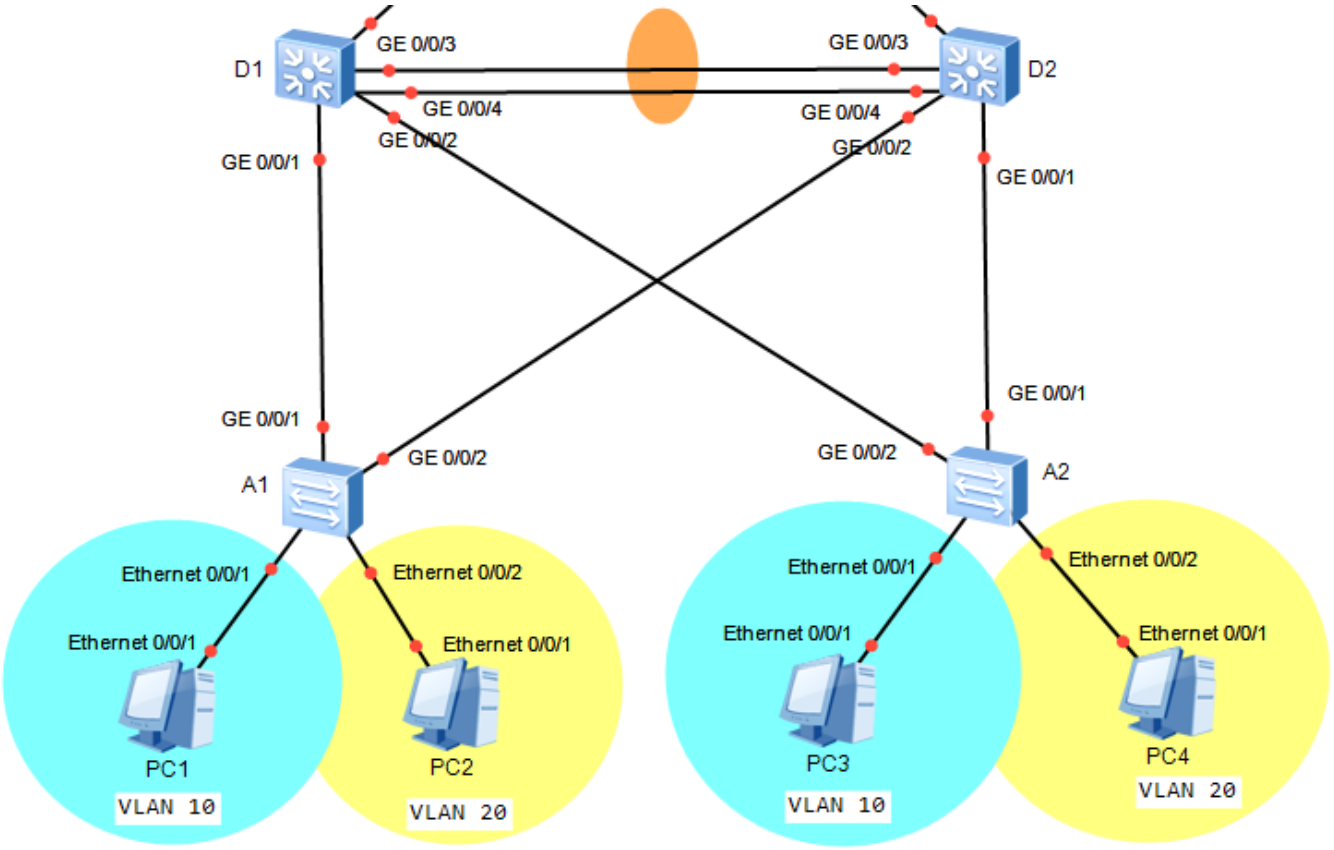
Huawei eNSP симуляторын қолданған жағдайда

Тақырыбы: Корпоративті (Enterprise) желі құру
Жұмыстың мақсаты: HCIA-Datacom деңгейін меңгеру
Жұмыстың орындалу қадамы:

1) VLAN құру және Access, Trunk port
   Link Aggregation. LACP
   MSTP (Multiple Spanning Tree Protocol)
2) VRRP (Virtual Router Redundancy Protocol)
3) OSPF
4) DHCP
5) NAT (Easy IP)
6) DNS және HTTP



1-қадам: Link Aggregation, Trunk/Access, MSTP конфигурациялау



--- D1, D2 ---

undo terminal monitor
system-view
sysname D1

vlan batch 10 20
display vlan

interface Eth-Trunk 1
port link-type trunk
port trunk allow-pass vlan 10 20
mode lacp-static

display port vlan

```
Port                    Link Type     PVID  Trunk VLAN List
--------------------------------------------------------------
Eth-Trunk1              trunk         1     10 20
GigabitEthernet0/0/1    hybrid        1     -
GigabitEthernet0/0/2    hybrid        1     -
```

int g0/0/3
shutdown
display this
eth-trunk 1

undo shutdown // бұл команданы 10 second-тан кейін орнындаймыз!

int g0/0/4
shutdown
display this
eth-trunk 1

undo shutdown // бұл команданы 10 second-тан кейін орнындаймыз!

display int brief

```
InUti/OutUti: input utility/output utility
Interface                   PHY   Protocol InUti OutUti   inErrors  outErrors
Eth-Trunk1                  up    up          0%     0%          0          0
  GigabitEthernet0/0/3      up    up          0%     0%          0          0
  GigabitEthernet0/0/4      up    up          0%     0%          0          0
GigabitEthernet0/0/1        up    up          0%     0%          0          0
GigabitEthernet0/0/2        up    up          0%     0%          0          0
GigabitEthernet0/0/5        down  down        0%     0%          0          0
```

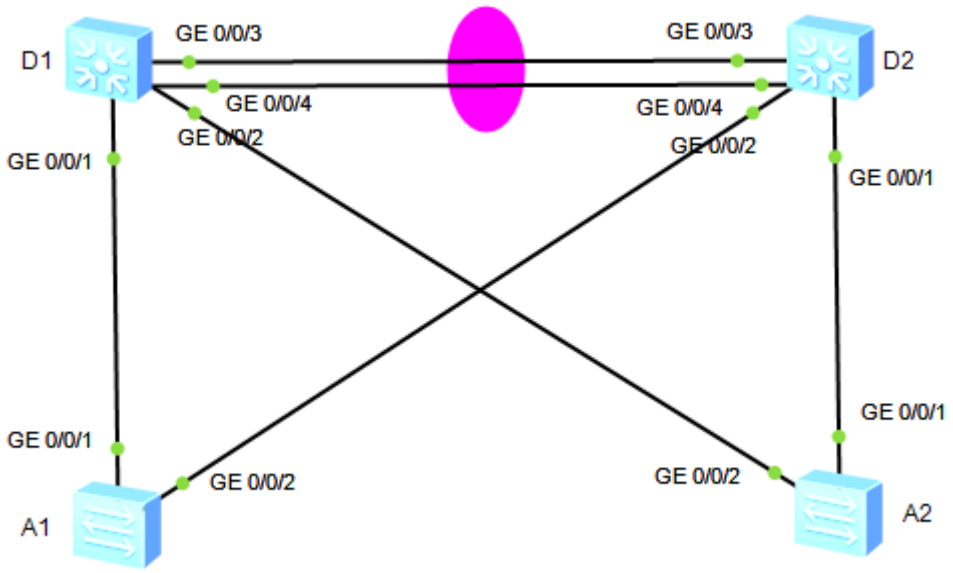
display eth-trunk 1

```
Eth-Trunk1's state information is:
Local:
LAG ID: 1                   WorkingMode: STATIC
Preempt Delay: Disabled     Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768      System ID: 4c1f-ccfe-137f
Least Active-linknumber: 1  Max Active-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/3   Selected 1GE      32768   4      305     10111100  1
GigabitEthernet0/0/4   Selected 1GE      32768   5      305     10111100  1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/3   32768    4c1f-cc11-6e84  32768   4      305     10111100
GigabitEthernet0/0/4   32768    4c1f-cc11-6e84  32768   5      305     10111100
```

display int eth-trunk 1

```
Eth-Trunk1 current state : UP
Line protocol current state : UP
Description:
Switch Port, PVID :    1, Hash arithmetic : According to SIP-XOR-DIP,Maximal BW:
 2G, Current BW: 2G, The Maximum Frame Length is 9216
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 4c1f-ccfe-137f
Current system time: 2023-03-22 15:15:33-08:00
    Input bandwidth utilization  :    0%
    Output bandwidth utilization :    0%
--------------------------------------------------
PortName                      Status      Weight
--------------------------------------------------
GigabitEthernet0/0/3          UP          1
GigabitEthernet0/0/4          UP          1
--------------------------------------------------
The Number of Ports in Trunk : 2
The Number of UP Ports in Trunk : 2
```



port-group group-member g0/0/1 to g0/0/2
port link-type trunk
port trunk allow-pass vlan 10 20

display port vlan

```
Port                      Link Type     PVID  Trunk VLAN List
--------------------------------------------------------------
Eth-Trunk1                trunk         1     1 10 20
GigabitEthernet0/0/1      trunk         1     1 10 20
GigabitEthernet0/0/2      trunk         1     1 10 20
GigabitEthernet0/0/3      hybrid        0     -
GigabitEthernet0/0/4      hybrid        0     -
```

display stp

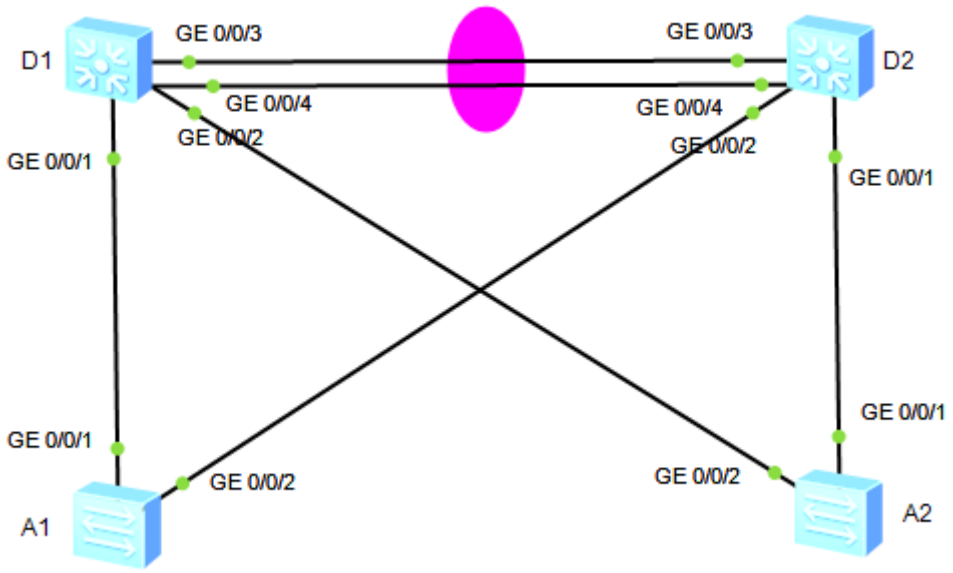
stp region-configuration
region-name LAN1
instance 1 vlan 10
instance 2 vlan 20
active region-configuration
quit

--- D1 ---

stp instance 1 priority 4096
stp instance 2 priority 8192

--- D2 ---

stp instance 1 priority 8192
stp instance 2 priority 4096



--- A1 ---

```
undo terminal monitor
system-view
sysname A1

vlan batch 10 20
display vlan

port-group group-member g0/0/1 to g0/0/2
port link-type trunk
port trunk allow-pass vlan 10 20

display port vlan

stp region-configuration
region-name LAN1
instance 1 vlan 10
instance 2 vlan 20
active region-configuration
quit

int g0/0/1
stp instance 2 cost 100000

int g0/0/2
stp instance 1 cost 100000
```

--- A2 ---

```
undo terminal monitor
system-view
sysname A2

vlan batch 10 20
```

display vlan

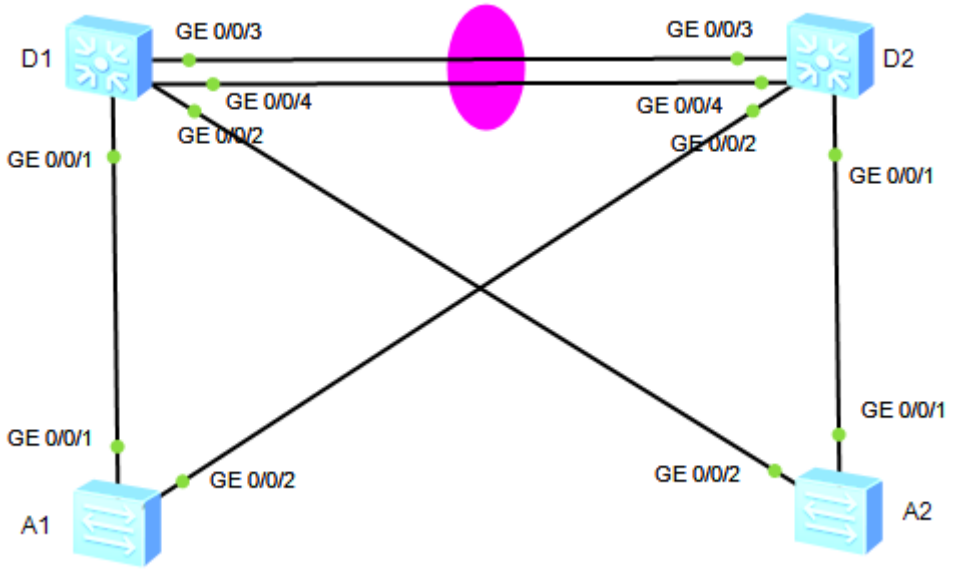
port-group group-member g0/0/1 to g0/0/2
port link-type trunk
port trunk allow-pass vlan 10 20

display port vlan

stp region-configuration
region-name LAN1
instance 1 vlan 10
instance 2 vlan 20
active region-configuration
quit

int g0/0/1
stp instance 1 cost 100000

int g0/0/2
stp instance 2 cost 100000



--- A1 ---

display stp instance 1 brief

```
MSTID  Port                      Role  STP State     Protection
  1    GigabitEthernet0/0/1      ROOT  FORWARDING      NONE
  1    GigabitEthernet0/0/2      ALTE  DISCARDING      NONE
```

display stp instance 2 brief

```
MSTID  Port                      Role  STP State     Protection
  2    GigabitEthernet0/0/1      ALTE  DISCARDING      NONE
  2    GigabitEthernet0/0/2      ROOT  FORWARDING      NONE
```

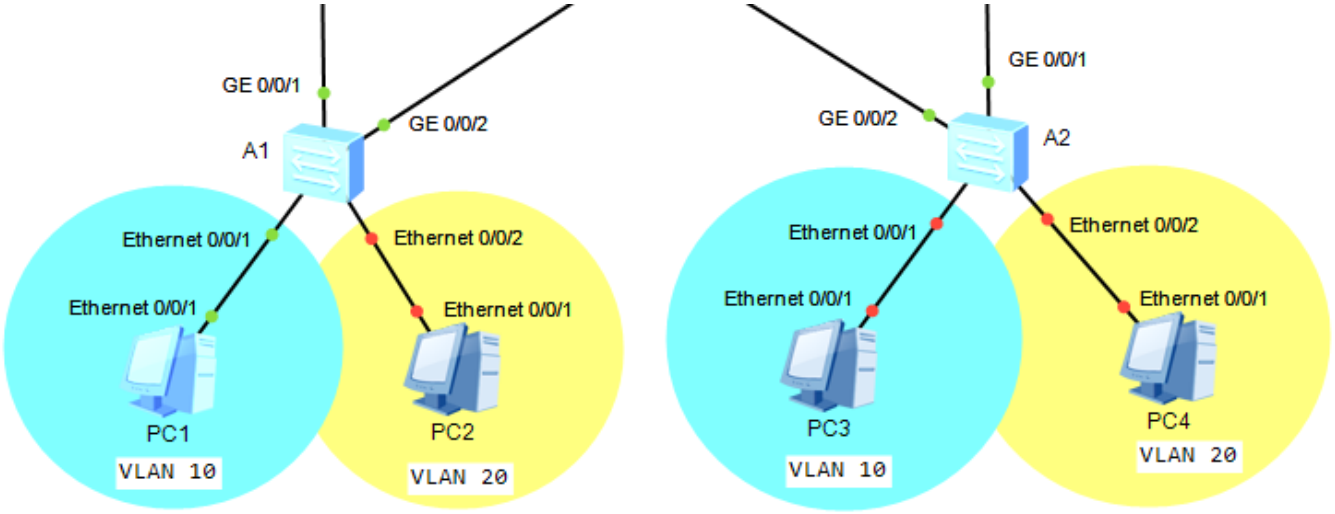
--- A2 ---

display stp instance 1 brief

```
MSTID  Port                      Role  STP State     Protection
  1    GigabitEthernet0/0/1      ALTE  DISCARDING      NONE
  1    GigabitEthernet0/0/2      ROOT  FORWARDING      NONE
```

display stp instance 2 brief

```
MSTID  Port                      Role  STP State     Protection
  2    GigabitEthernet0/0/1      ROOT  FORWARDING      NONE
  2    GigabitEthernet0/0/2      ALTE  DISCARDING      NONE
```



--- A1 ---

int Ethernet0/0/1
port link-type access
port default vlan 10

int Ethernet0/0/2
port link-type access
port default vlan 20

display vlan

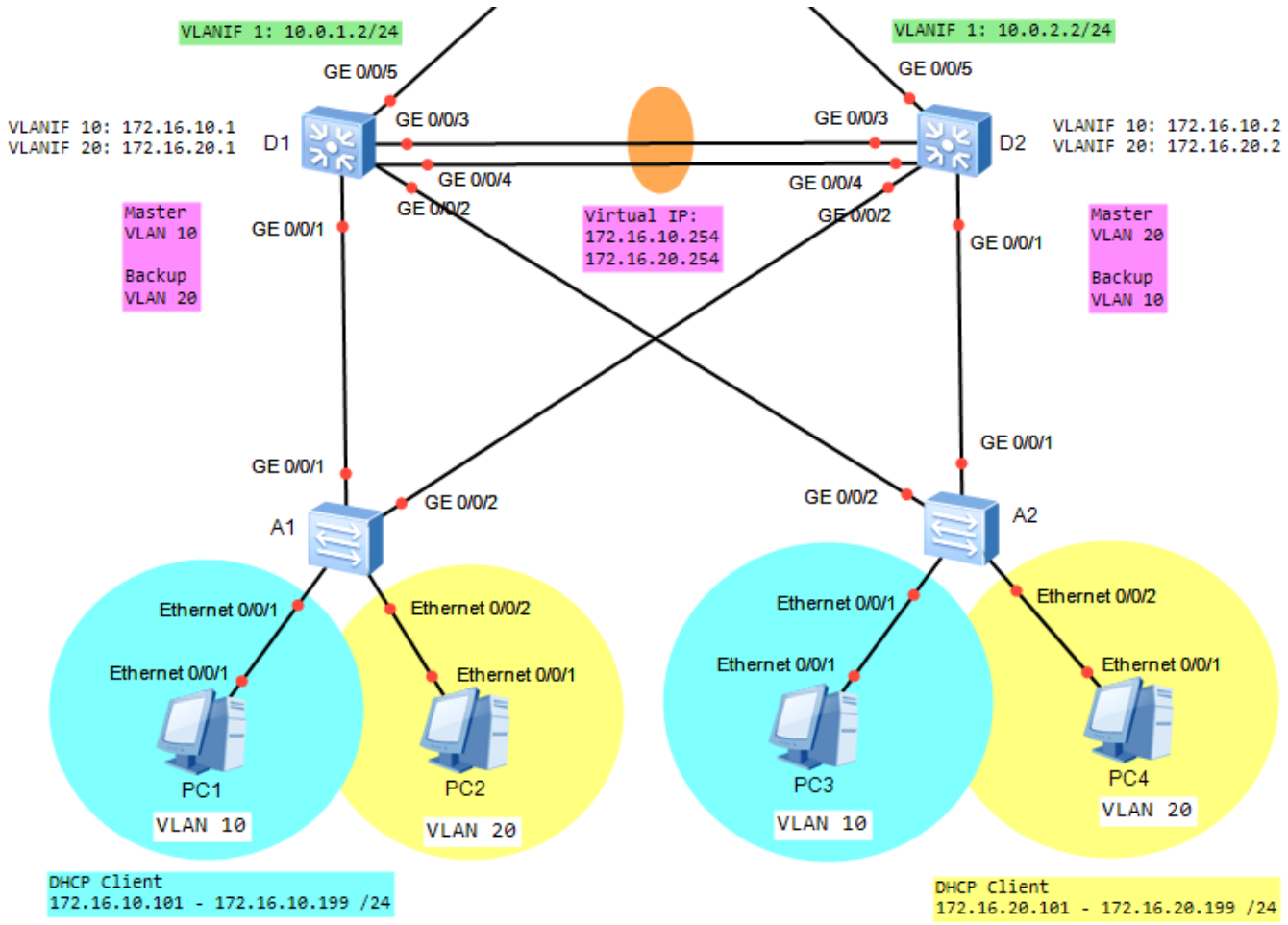
--- A2 ---

int Ethernet0/0/1
port link-type access
port default vlan 10

int Ethernet0/0/2
port link-type access
port default vlan 20

display vlan

2-қадам: VRRP (Virtual Router Redundancy Protocol) конфигурациялау



--- D1 ---

int vlanif 10
ip addr 172.16.10.1 24
vrrp vrid 1 virtual-ip 172.16.10.254
vrrp vrid 1 priority 110

int vlanif 20
ip addr 172.16.20.1 24
vrrp vrid 2 virtual-ip 172.16.20.254

display ip int brief

```
Vlanif10                    172.16.10.1/24      up          up
Vlanif20                    172.16.20.1/24      up          up
```

display vrrp brief

--- D2 ---

int vlanif 10
ip addr 172.16.10.2 24
vrrp vrid 1 virtual-ip 172.16.10.254

int vlanif 20
ip addr 172.16.20.2 24
vrrp vrid 2 virtual-ip 172.16.20.254

vrrp vrid 2 priority 110

[D2] display ip int brief

```
Vlanif10                      172.16.10.2/24       up         up
Vlanif20                      172.16.20.2/24       up         up
```

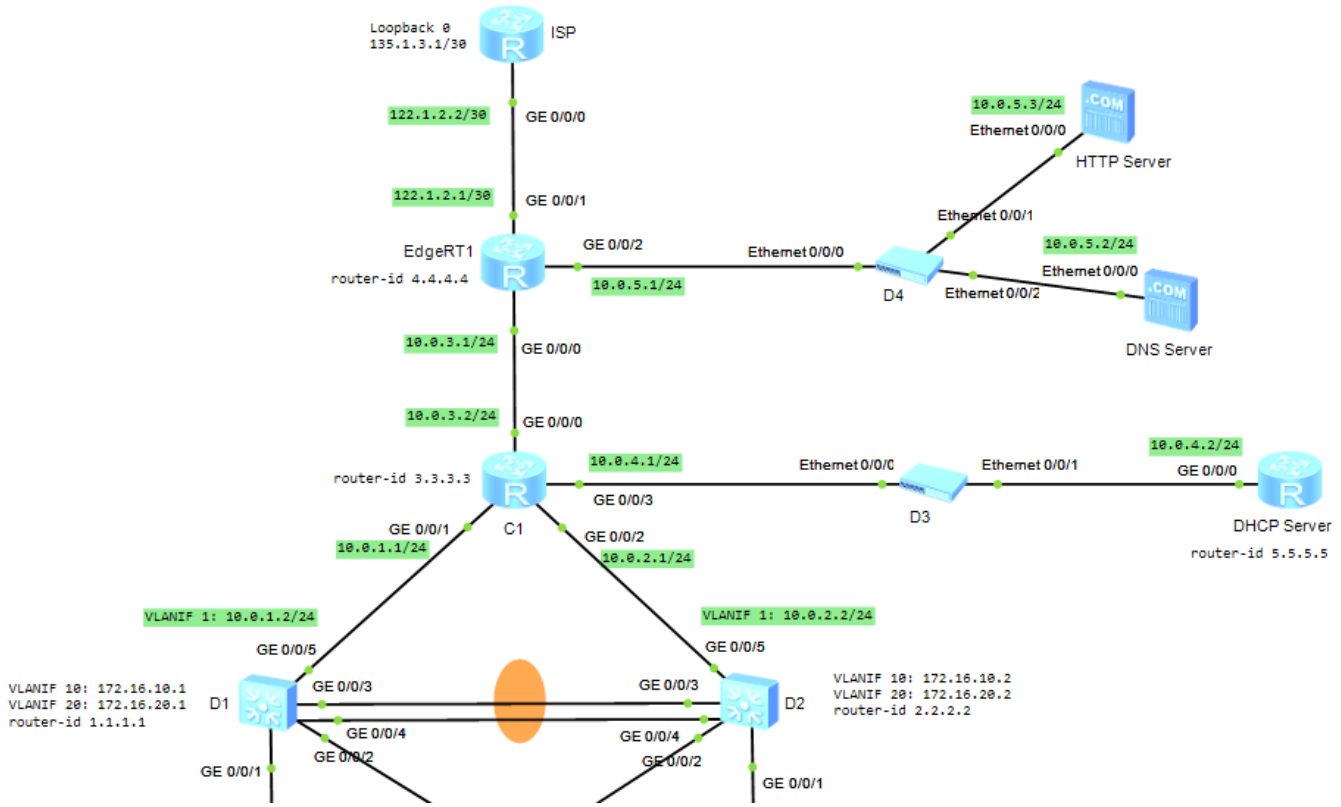
[D2] display vrrp brief

```
VRID  State        Interface                Type     Virtual IP
------------------------------------------------------------------
1     Backup       Vlanif10                 Normal   172.16.10.254
2     Master       Vlanif20                 Normal   172.16.20.254
------------------------------------------------------------------
Total:2     Master:1     Backup:1     Non-active:0
```

[D1] display vrrp brief

```
VRID  State        Interface                Type     Virtual IP
------------------------------------------------------------------
1     Master       Vlanif10                 Normal   172.16.10.254
2     Backup       Vlanif20                 Normal   172.16.20.254
------------------------------------------------------------------
Total:2     Master:1     Backup:1     Non-active:0
```

3-қадам: OSPF конфигурациялау



--- D1 ---

int vlanif 1
ip address 10.0.1.2 24

display ip int brief

```
Vlanif1                         10.0.1.2/24          up          up
Vlanif10                        172.16.10.1/24       up          up
Vlanif20                        172.16.20.1/24       up          up
```

ospf 1 router-id 1.1.1.1
area 0
network 10.0.1.0 0.0.0.255
network 172.16.10.0 0.0.0.255
network 172.16.20.0 0.0.0.255

display cu | begin ospf

--- D2 ---

int vlanif 1
ip address 10.0.2.2 24

display ip int brief

```
Vlanif1                         10.0.2.2/24          up          up
Vlanif10                        172.16.10.2/24       up          up
Vlanif20                        172.16.20.2/24       up          up
```

ospf 1 router-id 2.2.2.2
area 0
network 10.0.2.0 0.0.0.255
network 172.16.10.0 0.0.0.255
network 172.16.20.0 0.0.0.255

display ospf peer

--- C1 ---

undo terminal monitor
system-view
sysname C1

int g0/0/0
ip address 10.0.3.2 24
int g0/0/1
ip address 10.0.1.1 24
int g0/0/2
ip address 10.0.2.1 24
int g0/0/3
ip address 10.0.4.1 24

display ip int brief

ospf 1 router-id 3.3.3.3
area 0
network 10.0.1.0 0.0.0.255
network 10.0.2.0 0.0.0.255
network 10.0.3.0 0.0.0.255

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```
network 10.0.4.0 0.0.0.255

display ospf peer
```

--- EdgeRT1 ---

```
undo terminal monitor
system-view
sysname EdgeRT1

int g0/0/0
ip address 10.0.3.1 24
int g0/0/2
ip address 10.0.5.1 24
int g0/0/1
ip address 122.1.2.1 30

display ip int brief

ospf 1 router-id 4.4.4.4
area 0
network 10.0.3.0 0.0.0.255
network 10.0.5.0 0.0.0.255
network 122.1.2.0 0.0.0.3

display ospf peer
```

--- ISP ---

```
undo terminal monitor
system-view
sysname ISP

int g0/0/0
ip address 122.1.2.2 30
int LoopBack 0
ip address 135.1.3.1 30

display ip int brief
```

--- DHCP Server ---

```
undo terminal monitor
system-view
sysname DHCP

int g0/0/0
ip address 10.0.4.2 24

display ip int brief

ospf 1 router-id 5.5.5.5
```

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```
area 0
network 10.0.4.0 0.0.0.255

display ospf peer
```

4-қадам: DHCP хаттамасын конфигурациялау

```
--- DHCP Server ---

dhcp enable

ip pool VLAN10
network 172.16.10.0 mask 24
gateway-list 172.16.10.254
dns-list 10.0.5.2
excluded-ip-address 172.16.10.1 172.16.10.100
excluded-ip-address 172.16.10.200 172.16.10.253
lease day 5

ip pool VLAN20
network 172.16.20.0 mask 24
gateway-list 172.16.20.254
dns-list 10.0.5.2
excluded-ip-address 172.16.20.1 172.16.20.100
excluded-ip-address 172.16.20.200 172.16.20.253
lease day 5

interface g0/0/0
dhcp select global

display ip pool
display ip pool name VLAN10
display dhcp server statistics

--- D1 and D2 ---

dhcp enable

int vlanif 10
dhcp select relay
dhcp relay server-ip 10.0.4.2

int vlanif 20
dhcp select relay
dhcp relay server-ip 10.0.4.2

--- DHCP Client ---
```

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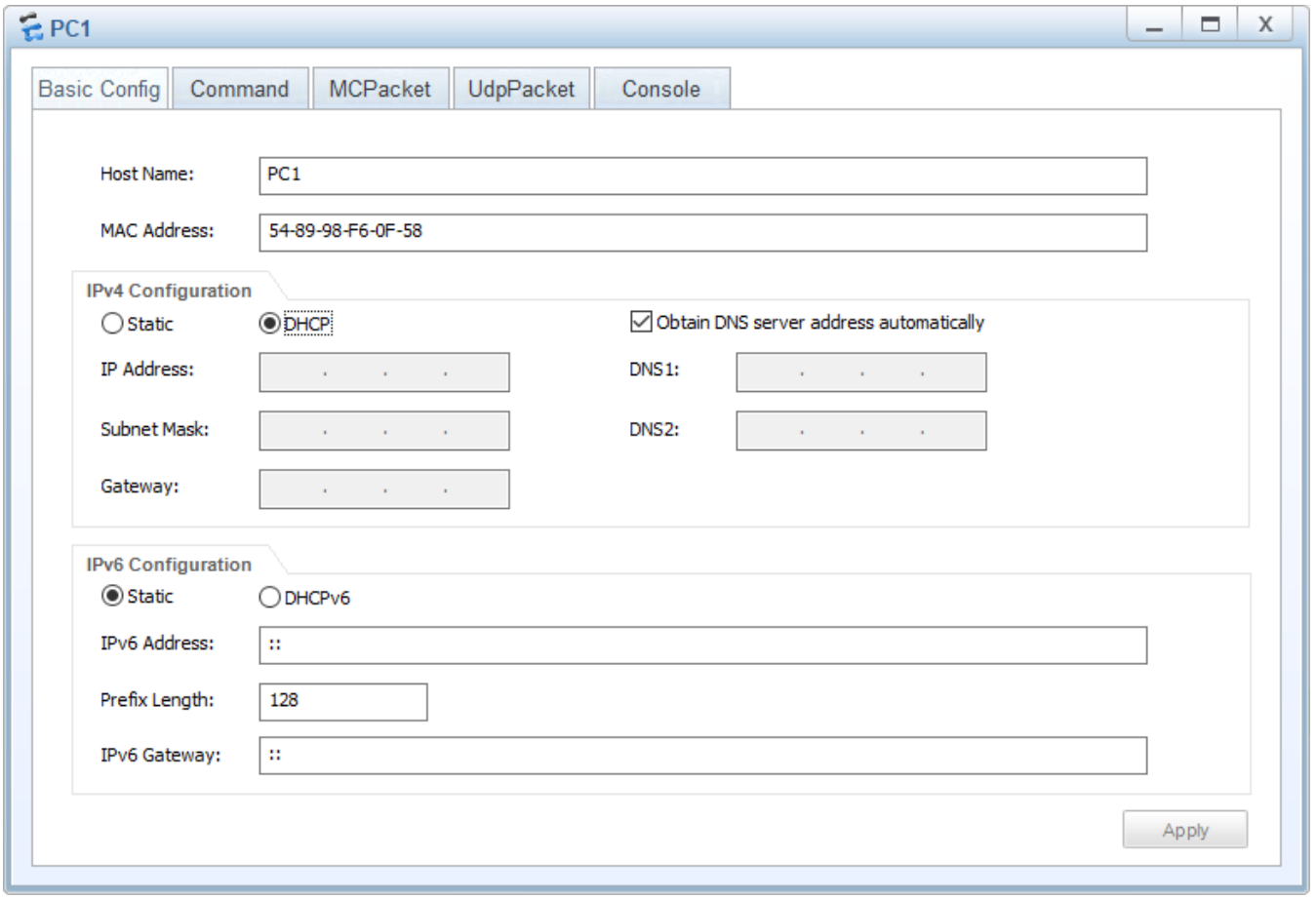


PC1

```
PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fef6:f58
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 172.16.10.199
Subnet mask.......................: 255.255.255.0
Gateway...........................: 172.16.10.254
Physical address..................: 54-89-98-F6-0F-58
DNS server........................: 10.0.5.2
```

PC2

```
PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fe40:2a62
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 172.16.20.198
Subnet mask.......................: 255.255.255.0
Gateway...........................: 172.16.20.254
Physical address..................: 54-89-98-40-2A-62
DNS server........................: 10.0.5.2
```

PC1

```
PC>ping 172.16.20.198

Ping 172.16.20.198: 32 data bytes, Press Ctrl_C to break
From 172.16.20.198: bytes=32 seq=1 ttl=127 time=94 ms
From 172.16.20.198: bytes=32 seq=2 ttl=127 time=109 ms
From 172.16.20.198: bytes=32 seq=3 ttl=127 time=94 ms
From 172.16.20.198: bytes=32 seq=4 ttl=127 time=109 ms
From 172.16.20.198: bytes=32 seq=5 ttl=127 time=125 ms
```

--- DHCP Server ---

display ip pool
display ip pool name VLAN10
display dhcp server statistics

5-қадам: NAT (Easy IP) конфигурациялау

--- EdgeRT1 ---

acl 2000
rule permit source 172.16.10.0 0.0.0.255
rule permit source 172.16.20.0 0.0.0.255

int g0/0/1
nat outbound 2000

display nat outbound

PC1

```
PC>ping 122.1.2.2

Ping 122.1.2.2: 32 data bytes, Press Ctrl_C to break
From 122.1.2.2: bytes=32 seq=1 ttl=252 time=109 ms
From 122.1.2.2: bytes=32 seq=2 ttl=252 time=94 ms
From 122.1.2.2: bytes=32 seq=3 ttl=252 time=94 ms
From 122.1.2.2: bytes=32 seq=4 ttl=252 time=125 ms
From 122.1.2.2: bytes=32 seq=5 ttl=252 time=93 ms
```

[EdgeRT1] display nat session all verbose

```
NAT Session Table Information:

    Protocol           : ICMP(1)
    SrcAddr   Vpn      : 172.16.10.199
    DestAddr  Vpn      : 122.1.2.2
    Type Code IcmpId   : 0   8   12249
    Time To Live       : 20 s
    NAT-Info
      New SrcAddr      : 122.1.2.1
      New DestAddr     : ----
      New IcmpId       : 10249
```

PC1

```
PC>ping 135.1.3.1

Ping 135.1.3.1: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!
```

--- EdgeRT1 ---

Default Static Routing

```
ip route-static 0.0.0.0 0.0.0.0 122.1.2.2
```

default-information originate in OSPF

```
ospf 1
default-route-advertise
```

--- C1 ---

display ip routing-table
display ip routing-table protocol ospf
display ospf routing

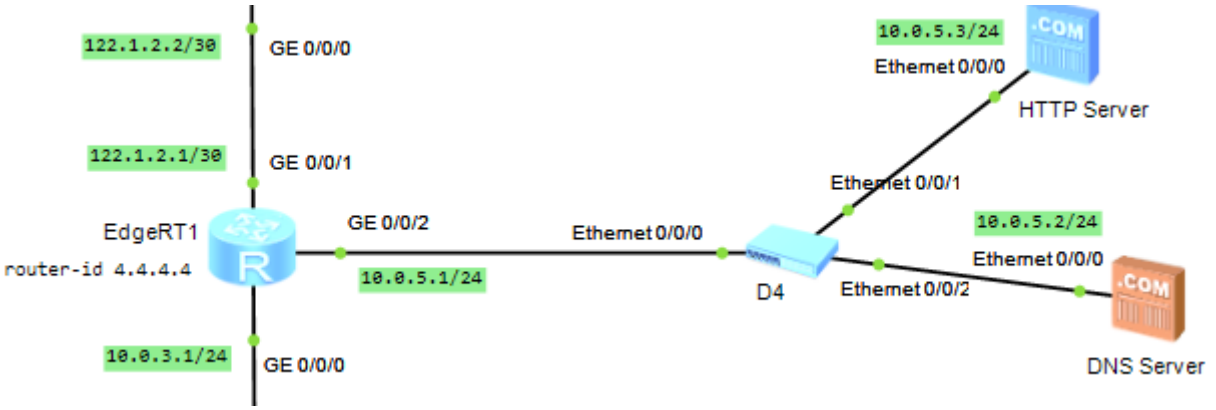
--- PC1 ---

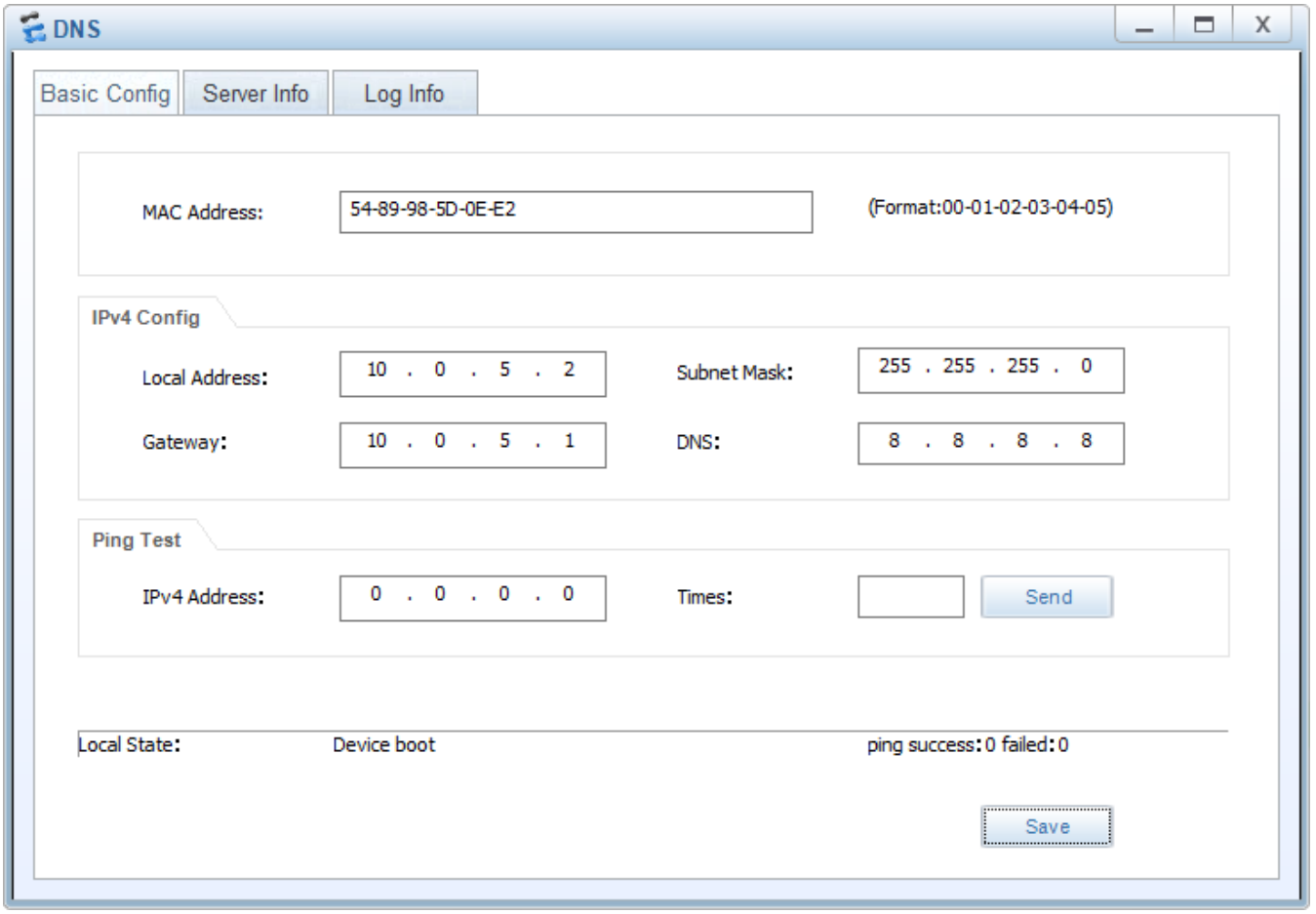
```
PC>ping 135.1.3.1

Ping 135.1.3.1: 32 data bytes, Press Ctrl_C to break
From 135.1.3.1: bytes=32 seq=1 ttl=252 time=94 ms
From 135.1.3.1: bytes=32 seq=2 ttl=252 time=125 ms
From 135.1.3.1: bytes=32 seq=3 ttl=252 time=94 ms
From 135.1.3.1: bytes=32 seq=4 ttl=252 time=94 ms
From 135.1.3.1: bytes=32 seq=5 ttl=252 time=62 ms
```

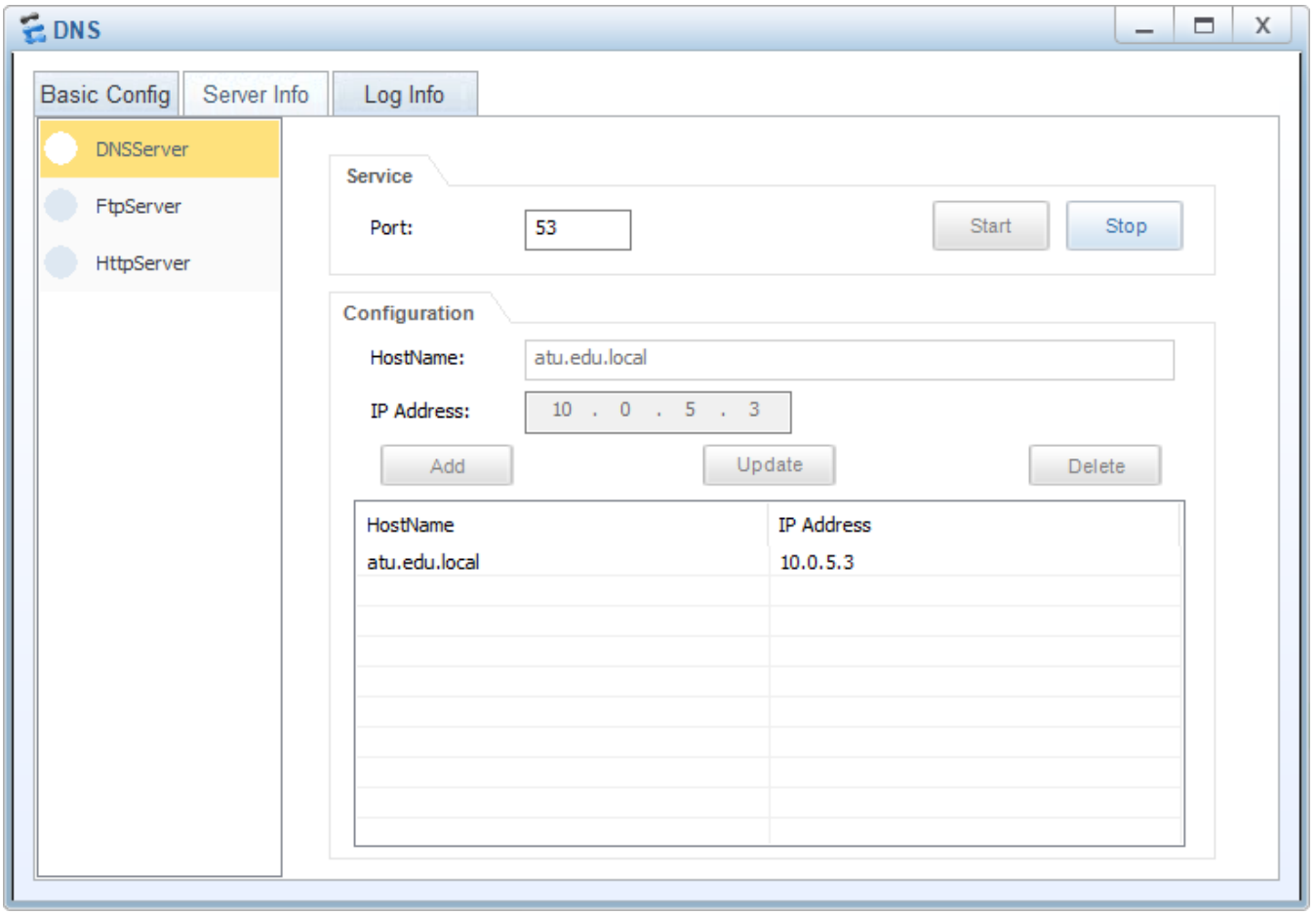
[EdgeRT1] display nat session all verbose

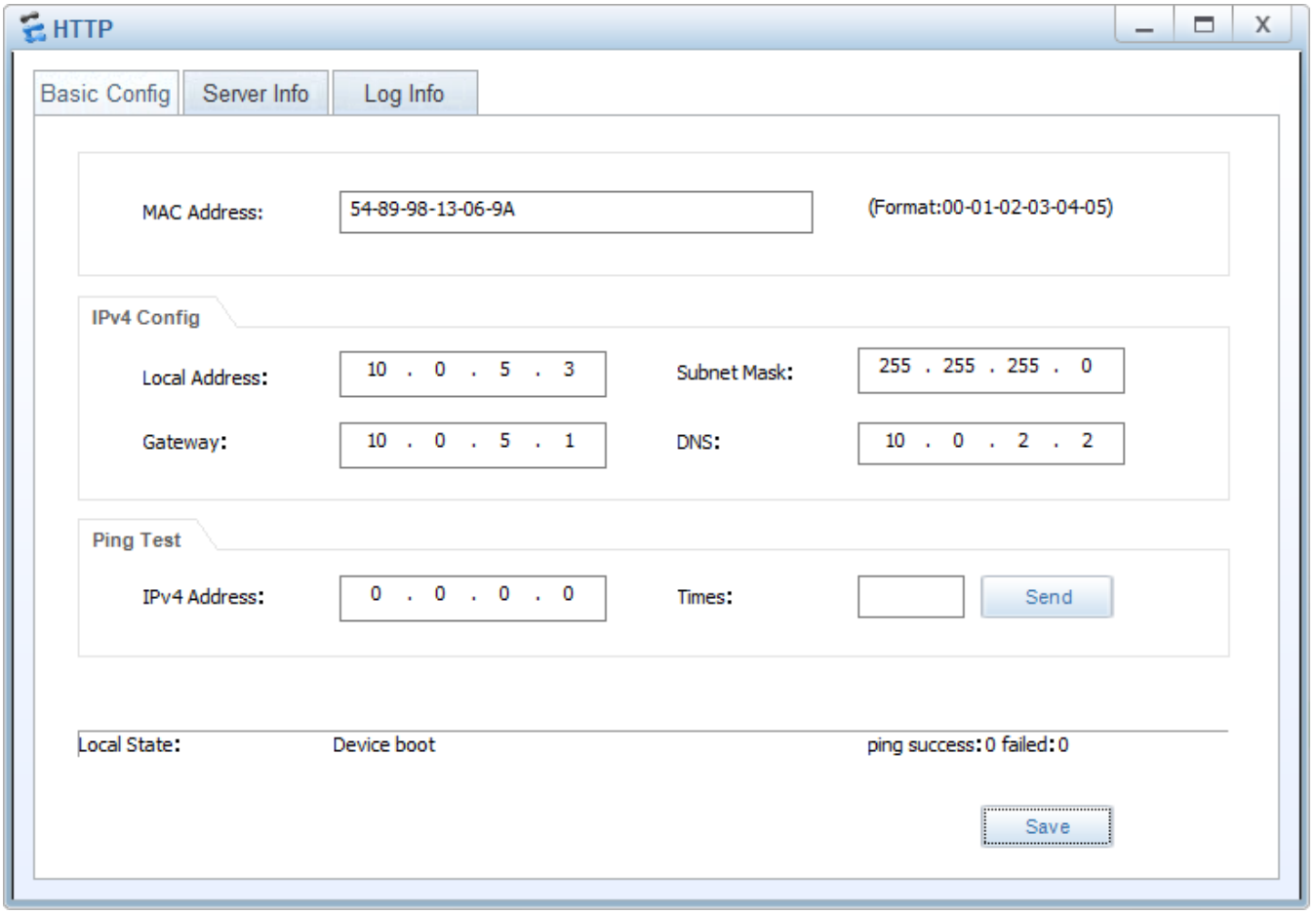
```
  Protocol           : ICMP(1)
  SrcAddr    Vpn     : 172.16.10.199
  DestAddr   Vpn     : 135.1.3.1
  Type Code IcmpId   : 0    8    31155
  Time To Live       : 20 s
  NAT-Info
    New SrcAddr      : 122.1.2.1
    New DestAddr     : ----
    New IcmpId       : 10254
```

6-қадам: HTTP және DNS конфигурациялау


122.1.2.2/30
GE 0/0/0
122.1.2.1/30
GE 0/0/1
EdgeRT1
router-id 4.4.4.4
GE 0/0/2
10.0.5.1/24
Ethernet 0/0/0
D4
10.0.5.3/24
Ethernet 0/0/0
HTTP Server
Ethernet 0/0/1
10.0.5.2/24
Ethernet 0/0/0
Ethernet 0/0/2
DNS Server
10.0.3.1/24
GE 0/0/0

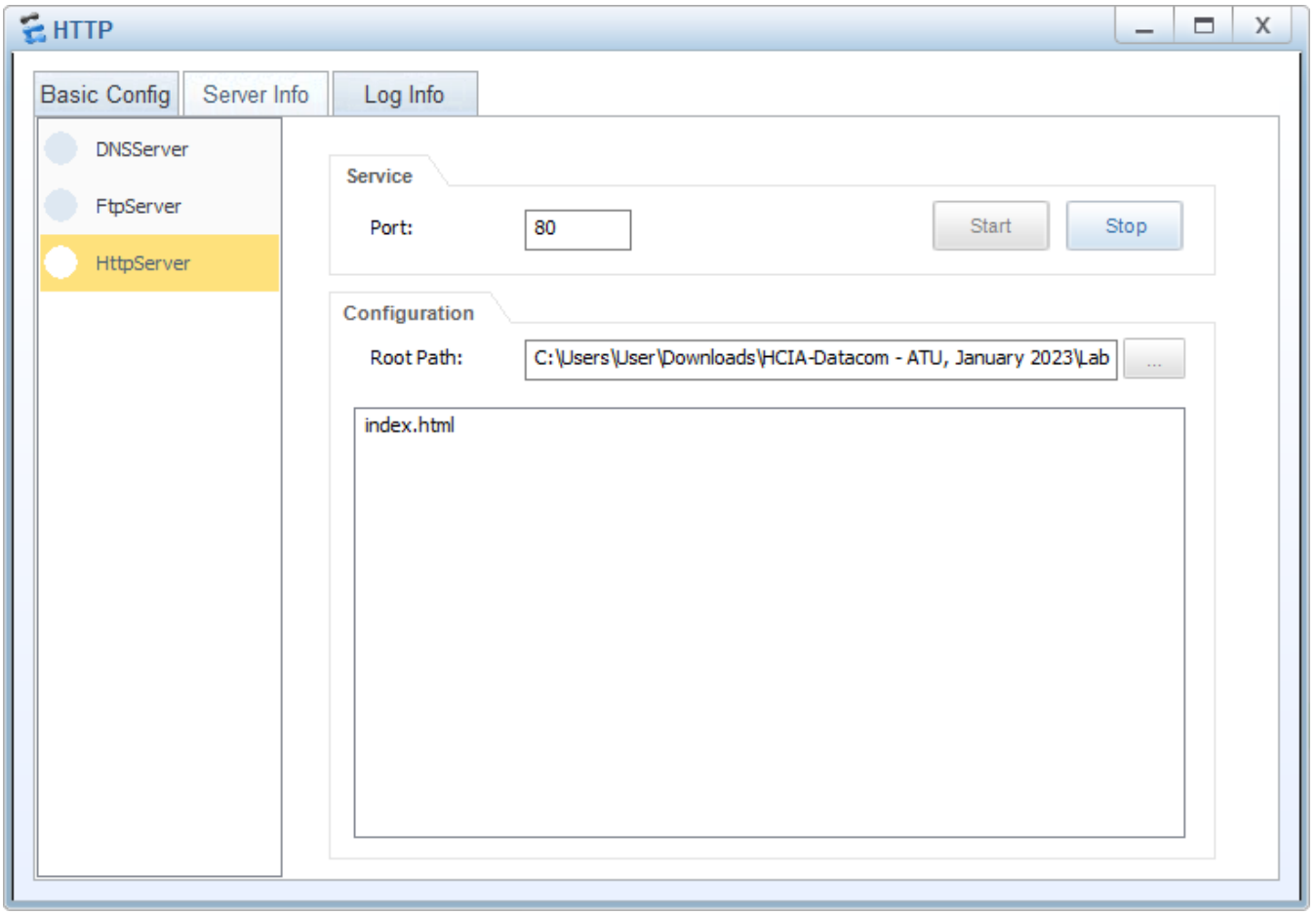

DNS
Basic Config
Server Info
Log Info
MAC Address:
54-89-98-5D-0E-E2
(Format:00-01-02-03-04-05)
IPv4 Config
Local Address:
10 . 0 . 5 . 2
Subnet Mask:
255 . 255 . 255 . 0
Gateway:
10 . 0 . 5 . 1
DNS:
8 . 8 . 8 . 8
Ping Test
IPv4 Address:
0 . 0 . 0 . 0
Times:
Send
Local State:
Device boot
ping success:0 failed:0
Save

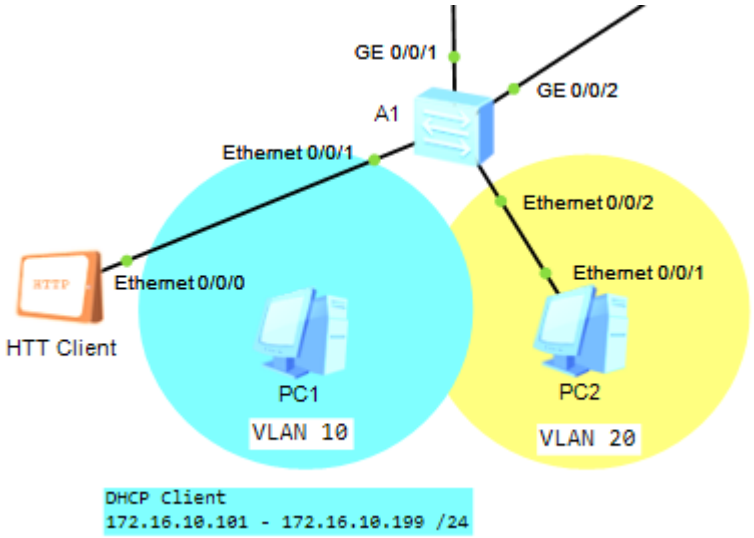

DNS
Basic Config
Server Info
Log Info
DNSServer
FtpServer
HttpServer
Service
Port:
53
Start
Stop
Configuration
HostName:
atu.edu.local
IP Address:
10 . 0 . 5 . 3
Add
Update
Delete
HostName
IP Address
atu.edu.local
10.0.5.3

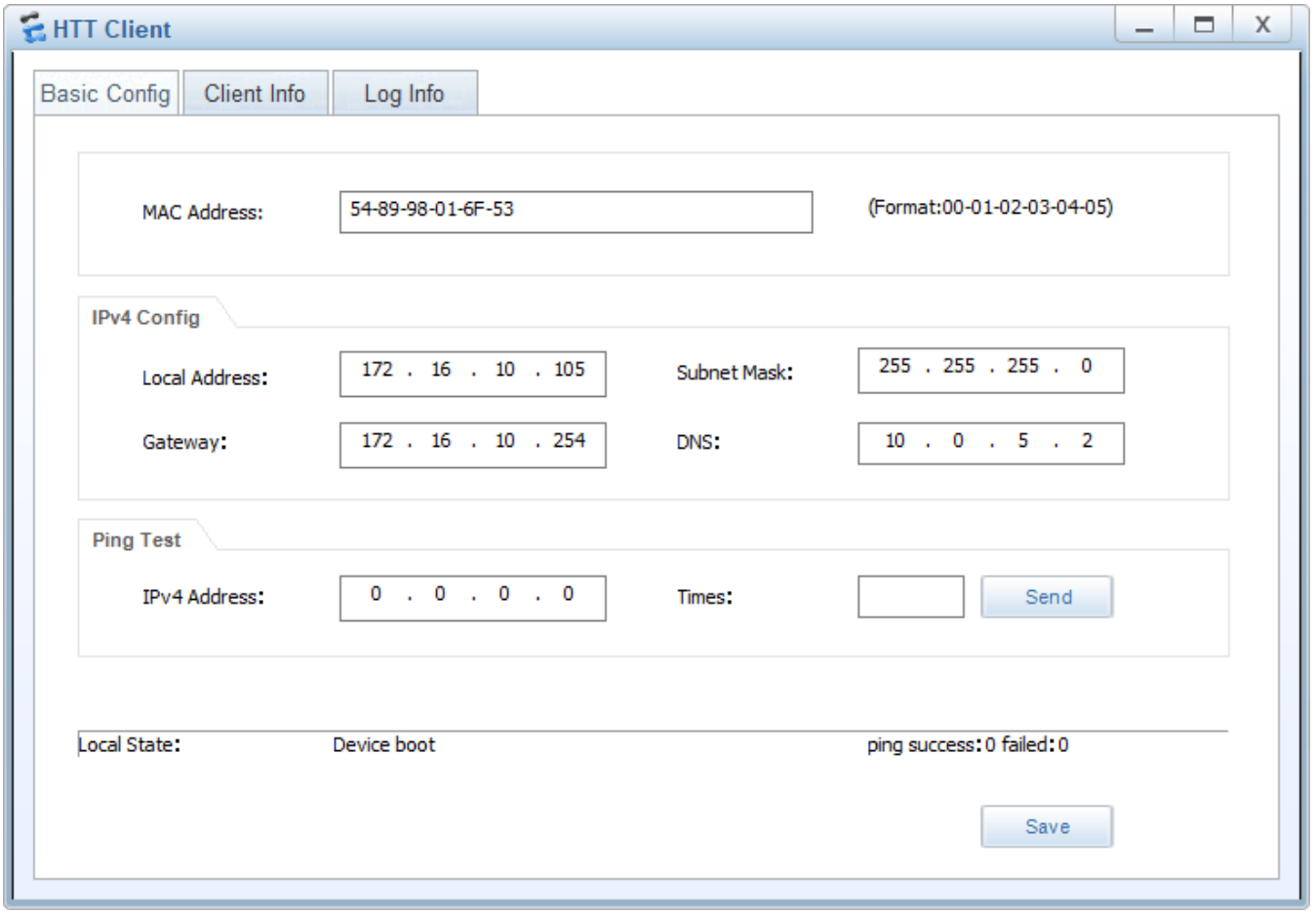

HTTP
Basic Config
Server Info
Log Info
MAC Address:
54-89-98-13-06-9A
(Format:00-01-02-03-04-05)
IPv4 Config
Local Address:
10 . 0 . 5 . 3
Subnet Mask:
255 . 255 . 255 . 0
Gateway:
10 . 0 . 5 . 1
DNS:
10 . 0 . 2 . 2
Ping Test
IPv4 Address:
0 . 0 . 0 . 0
Times:
Send
Local State:
Device boot
ping success:0 failed:0
Save

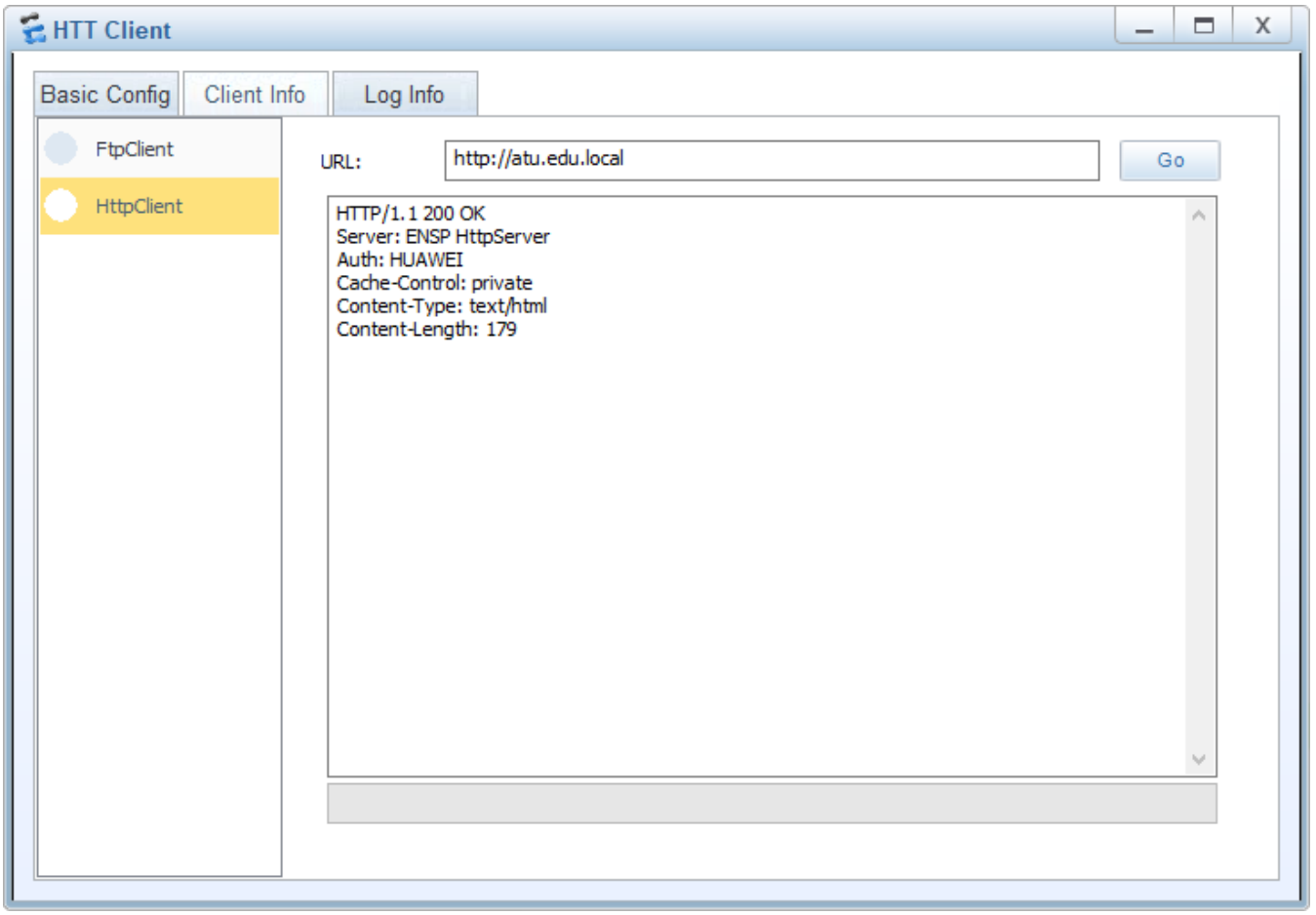
index.html

```
<!DOCTYPE html>
<html>
<head>
        <meta charset="UTF-8">
        <title>Example</title>
</head>
<body>
        <h1>Welcome to ATU!!!</h1>
</body>
</html>
```




GE 0/0/1
GE 0/0/2
A1
Ethernet 0/0/1
Ethernet 0/0/2
Ethernet 0/0/1
Ethernet 0/0/0
HTT Client
PC1
VLAN 10
PC2
VLAN 20
DHCP Client
172.16.10.101 - 172.16.10.199 /24


HTT Client
Basic Config
Client Info
Log Info
MAC Address:
54-89-98-01-6F-53
(Format:00-01-02-03-04-05)
IPv4 Config
Local Address:
172 . 16 . 10 . 105
Subnet Mask:
255 . 255 . 255 . 0
Gateway:
172 . 16 . 10 . 254
DNS:
10 . 0 . 5 . 2
Ping Test
IPv4 Address:
0 . 0 . 0 . 0
Times:
Send
Local State:
Device boot
ping success:0 failed:0
Save


HTT Client
Basic Config
Client Info
Log Info
FtpClient
HttpClient
URL:
http://atu.edu.local
Go
HTTP/1.1 200 OK
Server: ENSP HttpServer
Auth: HUAWEI
Cache-Control: private
Content-Type: text/html
Content-Length: 179

End ...